KRISHNA KUMAR TRIVEDI DS-EXP-3

```
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris

import warnings
warnings.filterwarnings('ignore')
```

```
iris=load_iris()
```

```
df=pd.DataFrame(iris.data,columns=iris.feature_names)
df["Species"]=iris.target
df["Species"]=df["Species"].map({
    0:"setosa",
    1:"versicolor",
    2:"virginica"

})
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with df New interactive sheet

```
sepal_length_report=df.groupby(["Species"])["sepal length (cm)"].agg(['mean', 'median', 'min', 'max', 'std'])
sepal_width_report=df.groupby(["Species"])["sepal width (cm)"].agg(['mean', 'median', 'min', 'max', 'std'])
petal_length_report=df.groupby(["Species"])["petal length (cm)"].agg(['mean', 'median', 'min', 'max', 'std'])
petal_width_report=df.groupby(["Species"])["petal width (cm)"].agg(['mean', 'median', 'min', 'max', 'std'])

print(f"Sepal length (cm) :\n{sepal_length_report} \n")
print(f"Sepal width (cm) :\n{sepal_length_report} \n")
print(f"Petal length (cm) :\n{sepal_length_report}\n")
print(f"Petal width (cm) :\n{sepal_length_report}\n")
```

```
Sepal length (cm) :
             mean  median  min  max       std
Species
setosa      5.006     5.0  4.3  5.8  0.352490
versicolor  5.936     5.9  4.9  7.0  0.516171
virginica   6.588     6.5  4.9  7.9  0.635880

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```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files datalarge.csv

- **datalarge.csv**(text/csv) - 1297 bytes, last modified: 3/6/2026 - 100% done

```
Saving datalarge.csv to datalarge.csv
```

```
data=pd.read_csv("datalarge.csv")
df=pd.DataFrame(data)
df.head()
```

	Gender	Age	Income
0	Female	45	73608
1	Female	33	52160
2	Male	33	48597
3	Female	26	74835
4	Male	40	75057

Next steps: Generate code with df New interactive sheet

```
gender_income_report=df.groupby(["Gender"])["Income"].agg(['mean', 'median', 'min', 'max', 'std'])
gender_age_report=df.groupby(["Gender"])["Age"].agg(['mean', 'median', 'min', 'max', 'std'])
print("Income grouped by gender:\n",gender_income_report,"\n")
print("Income grouped by gender:\n",gender_age_report,"\n")
```

```
Income grouped by gender:
            mean   median    min    max           std
Gender
Female  61526.7  62094.5  30267  86949  18220.210191
Male    56868.7  50235.0  30025  86568  18530.396289

Income grouped by gender:
           mean  median  min  max       std
Gender
Female  39.250    40.5   25   53  9.074478
Male    38.225    38.5   24   53  8.579903
```

Start coding or generate with AI.